

Escalate or Answer

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SoK: Agentic Retrieval-Augmented Generation (RAG): Taxonomy, Architectures, Evaluation, and Research Directions

Human-In-The-Loop Escalation Patterns

- ▶ A survey of agentic RAG architectures that names human-in-the-loop escalation as a callable action inside the agent's action space, triggered by epistemic uncertainty. Its non-escalating configuration is the baseline our System B adopts, and its escalation branch is the design pattern our System A instantiates.

“This pattern models human oversight as a callable API within the action space.”

MiPP: Minimal Proactivity Policies

How to escalate

- ▶ A contract for system-initiated interruptions: initiations must be bounded and reversible, must expose an explicit provenance cue naming the source that justified the pause, and must default to silence rather than speculation when evidence is insufficient. This is the shape of System A's escalation branch.

“Every initiation must expose an explicit provenance cue indicating the observable source(s) that justified the initiation ... enough local information for the user to judge whether the initiation is legitimate in context.”

“Trust Me, Bro!” On the Influence of Agentic Support on Trust Calibration in AI-Assisted Decision Making

What is trust

“Trust calibration denotes the alignment between users' trust and the system's underlying capability for a given task. Results show a positive direct effect of stated accuracy on trust and a significant interaction: accuracy shifts trust more strongly when reasoning is convincing.”

FictionalQA: A Dataset for Studying Memorization and Knowledge Acquisition

Uncontaminated corpus construction

- ▶ A five-stage pipeline (seed, fictsheets, fiction, Q&A, annotation) that generates fictional documents and certifies them by prompting the generator model blind and then informed, discarding items answerable without the document. Our corpus and contamination check follow this pipeline directly.

From Raw Corpora to Domain Benchmarks: Automated Evaluation of LLM Domain Expertise

Corpus contamination benchmarks

- ▶ Sharma et al.'s pipeline targets large scraped domain corpora. Our corpus is small, invented, and hand-authored, so we follow Kirchenbauer et al.'s blind-vs-informed check rather than their vocabulary-extraction approach.

Exploring Trust and Transparency in Retrieval-Augmented Generation for Domain Experts

What the people want!

- ▶ A study of 50 financial professionals using a RAG tool finds that confidence scores alone do not move trust, while source attribution does; open-ended responses show experts want control over which sources the system may draw from.

“Control Over Sources: ... 15 out of 50 participants highlighted the need to shortlist specific reports and trust certain sources, as evidenced by comments like, ‘I would personally trust some sources more than others.’ “

Plan Explanations as Model Reconciliation: An Empirical Study

The grading procedure

- ▶ A human-in-the-loop study where participants act as an external commander, evaluate plans produced by a robot, and rate them optimal or suboptimal, with end-of-study trust items.

Multimedia and Immersive Training Materials Influence Impressions of Learning but Not Learning Outcomes

Inflated confidence and trust

- ▶ Richer training materials raised participants' confidence that they would remember correctly, with no matching gain in actual ability to identify the next step. This dissociation motivates a task-embedded post-test instead of a self-report or declarative quiz.